24 October 2026

The Editor
Journal of Flood Risk Management

Dear Editor,

I am pleased to submit the manuscript “When do flood susceptibility models beat the flood record? Evidence from radar-observed floods in Bangladesh” for consideration as an Original Paper in the Journal of Flood Risk Management.

Every flood manager in a radar-mapped region already holds a free flood map, the record of where floods have been, yet published flood susceptibility maps are almost never tested against it. Weather and river forecasts must beat persistence before their skill is believed, while susceptibility studies in Bangladesh report areas under the ROC curve (AUC) of 0.87 to 0.98, usually from random splits of spatially clustered points that test a model beside its own training data. The manuscript sets a gradient-boosted terrain model against that record on later floods mapped from Sentinel-1 radar, scoring both on 10 km blocks held out from training, and asks when the record is enough and when a model adds to it.

Across 15 tests in seven regions, the record’s lead fell as the share of a flood reaching ground no earlier flood had reached rose (Spearman $\rho = -0.775$, $p < 0.001$), and the two were level at about a third of flooded assets on new ground (95 % interval 18–41 %, or 3–43 % when whole regions are resampled). The relation held with each region left out in turn and within every region with more than one test. Where floods returned to the same ground the record won (AUC 0.896 against 0.801 in Rangpur and Rajshahi) and kept its lead with masks re-mapped from a single satellite, with other training targets and with 40 km blocks, while on the cyclone-affected coast the model led. A global hydrodynamic hazard map was no stronger than the model.

The paper suits the Journal because its conclusions concern how flood risk is managed. It recommends that agencies publish the radar flood record as a planning baseline, and that the choice between record and model follow how much of past flooding fell on new ground. Its limits are stated plainly: scores measure agreement with flooding as radar sees it, 10 of the 15 tests cannot separate record from model, and all regions lie in one country. The code is at https://github.com/rakibhhridoy/rimes-mapathon and the data at https://doi.org/10.5281/zenodo.23573843.

The main text runs to just under 6,000 words with 5 figures and 7 tables, plus a Supporting Information file. The manuscript is original and not under consideration elsewhere. I am the sole author, declare an affiliation with Fermium Systems, which hosts the public dashboard, and received no specific funding.

Suggested reviewers:

- Dr Beth Tellman, Nelson Institute for Environmental Studies, University of Wisconsin–Madison, USA. Beth.tellman@wisc.edu
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Thank you for considering the manuscript.

Yours sincerely,

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